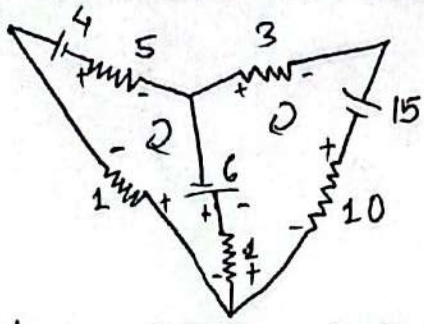
Aayush Bist
Roll No: 43
Group: CE-1-II (A)

1)
a)



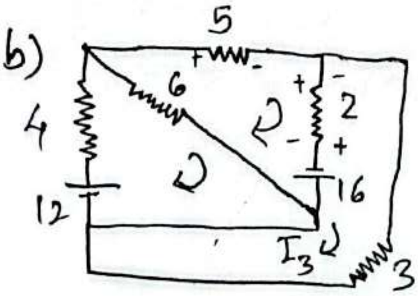
For Mesh 1, $(-7)I_1 + (1)I_2 = -2$ — (i)
for Mesh 2, $1I_1 + (-14)I_2 = 21$ — (ii)

From (i) & (ii),

$I_1 = 0.07\,A$
$I_2 = -1.5\,A$

∴ Current through $5\,\Omega$ resistor is 0.07 A right.

b)



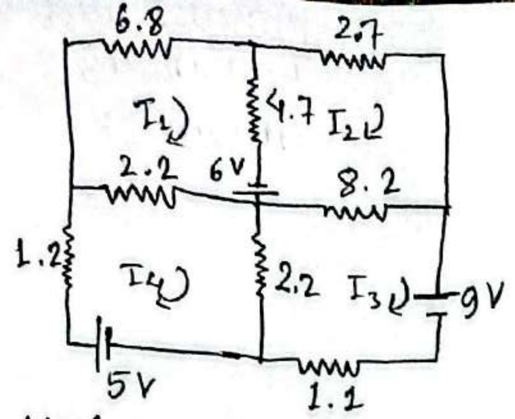
For mesh 1, $-10I_1 + 6I_2 + 0(I_3) = -12$ — (1)
For mesh 2, $6I_1 + (-13)I_2 + 2I_3 = -16$ — (2)
For mesh 3, $0I_1 + 2I_2 + 5I_3 = 16$ — (3)

solving (1), (2) & (3),

$I_1 = 2.94\,A$
$I_2 = 2.9\,A$
$I_3 = 2.04\,A$

∴ Current through $5\,\Omega$ resistor is 2.9 A right.

2 (a)



For Mesh 1, $-13.7I_1 + 4.7I_2 + 0I_3 + 2.2I_4 = -6$ — (1)
For Mesh 2, $4.7I_1 + (-15.6)I_2 + 8.2I_3 + 0I_4 = 6$ — (2)
For Mesh 3, $0I_1 + 8.2I_2 + (-31.3)I_3 + 22I_4 = 9$ — (3)
For Mesh 4, $2.2I_1 + 0I_2 + 22I_3 + (-25.4)I_4 = -5$ — (4)

On solving,

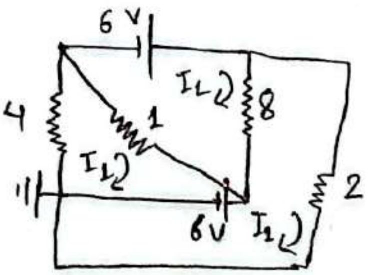
$I_1 = -0.032\ mA$

$I_2 = -0.883\ mA$

$I_3 = -0.968\ mA$

$I_4 = -0.639\ mA$

(b)



For Mesh 1, $-5I_1 + 1I_2 + 0I_3 = 6$ — (1)
For Mesh 2, $1I_1 - 9I_2 + 8I_3 = -6$ — (2)
For Mesh 3, $0I_1 + 8I_2 - 10I_3 = -6$ — (3)

On solving,

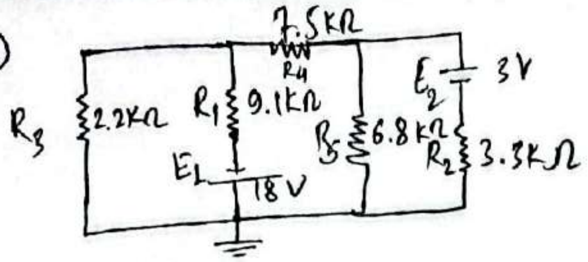
$I_1 = -3.8A$

$I_2 = -4.2A$

$I_3 = 0.2A$

(3)

a)



For mesh 1,

$-11.3I_1 + 9.1I_2 + 0I_3 = -18$ — ①

For Mesh 2,

$9.1(I_1) + (-23.4)I_2 + 6.8I_3 = 18$ — ②

For Mesh 3,

$0I_1 + 6.8I_2 - 10.1I_3 = 3$ — ③

Solving these.

$I_1 = 1.2mA$

$I_2 = -0.48mA$

$I_3 = -0.62mA$

$\therefore$ $2.2K\Omega \rightarrow 1.2mA$ up

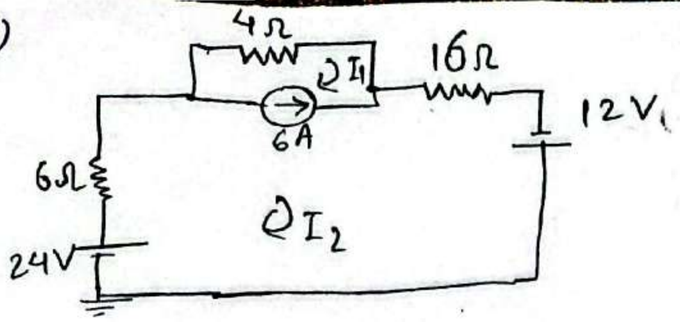
$3.3K\Omega \rightarrow 0.62mA$ up

$7.5K\Omega \rightarrow 0.48mA$ left

$9.1K\Omega \rightarrow 1.68mA$ down

$6.8K\Omega \rightarrow 0.14mA$ down

4(a)



$I_2 - I_1 = 6$

$-I_1 + 2I_2 = 6$ — (1)

Using supermesh,

$-4I_1 - 16I_2 = -24 - 12$

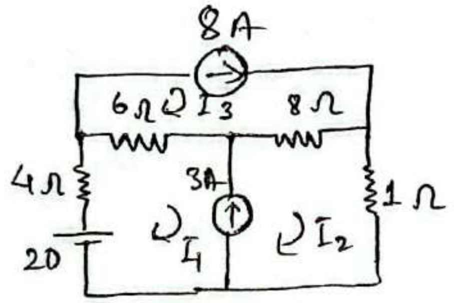
$-4I_1 - 16I_2 = -36$ — (2)

From (1) & (2),

$\therefore$ $I_1 = -3A$

$I_2 = 3A$

(b)



$I_3 = 8A$

Using supermesh,

$-10I_1 + 6I_3 - 9I_2 + 8I_3 = -20$

or, $-10I_1 - 9I_2 = -132$ — (1)

Now

$I_2 - I_1 = 3$

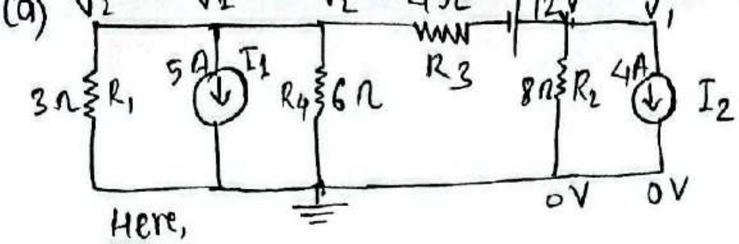
$-I_1 + I_2 = 3$ — (2)

Solving (1) & (2), we get

$I_1 = 5.52A$

$I_2 = 8.52A$

5 (a)



Here,

$V_1 - V_2 = 12$

$-V + V_1 + 0V_2 = 12$ — ①

using node analysis,

$I = \frac{V_1}{8} + 4 = \frac{-V_2}{6} - \frac{V_2}{3} - 5 = \frac{V_2 - V}{4}$

$I = \frac{V_1}{8} + 4 = \frac{-V_2}{2} - 5 = \frac{V_2 - V}{4}$

or, $\frac{1}{8}V_1 + \frac{1}{2}V_2 = -9$

or, $0V + \frac{1}{8}V_1 + \frac{1}{2}V_2 = -9$ — ②

And,

$\frac{1}{4}V + \frac{1}{8}V_1 - \frac{1}{4}V_2 = -4$ — ③

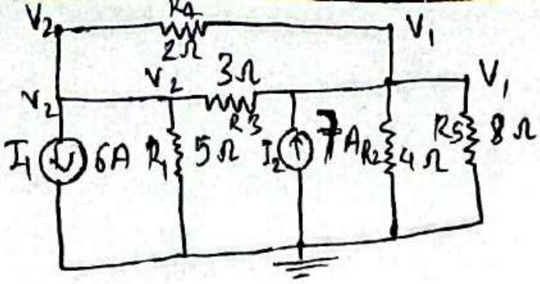
After solving ①, ② & ③,

$V = -24.57\,V$

$V_1 = -12.57\,V$

$V_2 = -14.85\,V$

5(b)



Here,

$$6 = -\frac{V_2}{5} + 7 - \frac{V_1}{4} - \frac{V_1}{8}$$

or, $\frac{3}{8}V_1 + \frac{1}{5}V_2 = 1$ — (1)

And,

$$7 - \frac{V_1}{4} - \frac{V_1}{8} = \frac{V_1}{3} - \frac{V_2}{3} + \frac{V_1}{2} - \frac{V_2}{2}$$

or, $7 = \frac{29}{24}V_1 - \frac{5}{6}V_2$ — (2)

Solving these,

$V_1 = 4.03V$

$V_2 = -2.55$ V

$2\Omega \rightarrow 6.58$ V

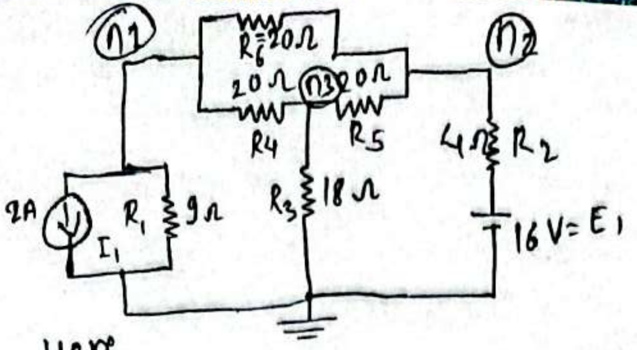
$3\Omega \rightarrow 6.58$ V

$4\Omega \rightarrow 4.03$ V

$5\Omega \rightarrow -2.55$ V

$8\Omega \rightarrow 4.03$ V

6(b)



Here,

$$I = \frac{V_2 - V_1}{20} + \frac{V_2 - V_3}{20} = \frac{V_3}{18} + 2 + \frac{V_1}{9} = \frac{16 - V_2}{4}$$

$$\text{or, } -\frac{V_1}{20} + \frac{V_2}{10} - \frac{V_3}{20} = \frac{V_1}{9} + \frac{V_3}{18} + 2 = 4 - \frac{V_2}{4}$$

$$\left(-\frac{1}{20}\right)V_1 + \left(\frac{7}{20}\right)V_2 - \left(\frac{1}{20}\right)V_3 = 4 \quad \text{— (1)}$$

$$\left(\frac{1}{9}\right)V_1 + \left(\frac{1}{4}\right)V_2 + \left(\frac{1}{18}\right)V_3 = 2 \quad \text{— (2)}$$

similarly,

$$\frac{V_2 - V_3}{20} = \frac{V_3}{18} + \frac{V_3 - V_1}{20}$$

$$\text{or, } \frac{1}{20}V_1 + \frac{1}{20}V_2 - \frac{7}{45}V_3 = 0 \quad \text{— (3)}$$

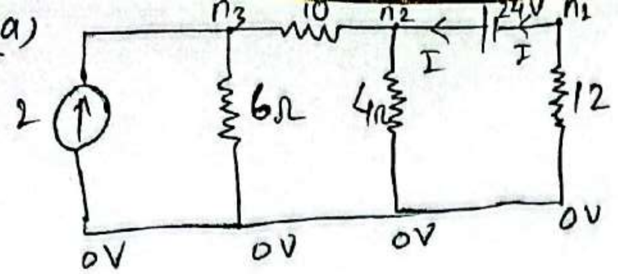
solving these we get,

$V_{node1} = -6.64\,V$

$V_{node2} = 10.66\,V$

$V_{node3} = 1.3\,V$

7(a)



Here,

$$\frac{V_2-V_3}{10} + \frac{V_2}{4} = \frac{-V_1}{12} = \frac{V_2}{4} + \frac{V_3}{6} - 2$$

$$\text{or, } \frac{V_2}{10} - \frac{V_3}{10} + \frac{V_2}{4} = \frac{-V_1}{12} = \frac{V_2}{4} + \frac{V_3}{6} - 2$$

$$\left(\frac{1}{12}\right) V_1 + \frac{7}{20} V_2 - \frac{1}{10} V_3 = 0 \quad \text{— (1)}$$

$$0V_1 + \frac{1}{10} V_2 - \frac{4}{15} V_3 = -2 \quad \text{— (2)}$$

Also,

$$V_2 - V_1 = 24$$

$$-1V_1 + V_2 + 0V_3 = 24 \quad \text{— (3)}$$

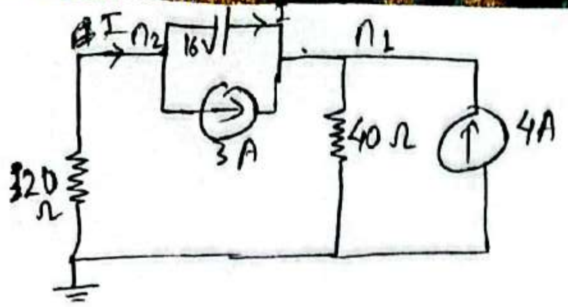
solving these we get,

$$V_{node\,1} = -17.05\ V$$

$$V_{node\,2} = 6.95\ V$$

$$V_{node\,3} = 10.1\ V$$

7 (b)



Here,

$$I + I_3 = \frac{V_1}{40} - 4$$

$$I = \frac{V_1}{40} - 7 \quad \text{—} (1)$$

And $$-\frac{V_2}{20} = I + 3$$

or, $$I = -\frac{V_2}{20} - 3 \quad \text{—} (2)$$

From (1) & (2)

$$\frac{V_1}{40} - 7 = -\frac{V_2}{20} - 3$$

or, $$\frac{1}{40} V_1 + \frac{1}{20} V_2 = 4 \quad \text{—} (3)$$

Ily,

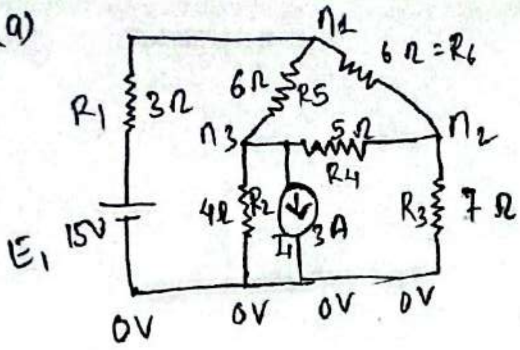
$$1V_1 + (-1) V_2 = 16 \quad \text{—} (4)$$

on solving (3) & (4), we get,

$V_{node1} = 64$ V

$V_{node2} = 48$ V

6(a)



Using node analysis,

$$\frac{V_1 - V_3}{6} + \frac{V_1 - V_2}{6} = \frac{V_3}{4} + 3 + \frac{V_2}{7} = \frac{15 - V_1}{3}$$

or, $$\frac{V_1}{3} - \frac{V_2}{6} - \frac{V_3}{6} = \frac{V_2}{7} + \frac{V_3}{7} + 3 = 5 - \frac{V_1}{3}$$

$$\left(\frac{2}{3}\right)V_1 + \left(-\frac{1}{6}\right)V_2 + \left(-\frac{1}{6}\right)V_3 = 5 \quad \text{— (1)}$$

$$\left(\frac{1}{3}\right)V_1 + \left(\frac{1}{7}\right)V_2 + \left(\frac{1}{7}\right)V_3 = 2 \quad \text{— (2)}$$

Similarelly,

$$\frac{V_1 - V_2}{6} = \frac{V_2}{7} + \frac{V_2 - V_3}{5}$$

$$\frac{V_1 - V_2}{6} - \frac{V_2}{7} - \frac{V_1}{5} + \frac{V_3}{5} = 0$$

$$\frac{1}{6}V_1 + \left(-\frac{107}{210}\right)V_2 + \left(\frac{1}{5}\right)V_3 = 0 \quad \text{— (3)}$$

Solving these we get,

$V_{node1} = 6.94\ V$

$V_{node2} = 1\ V$

$V_{node3} = -3.21\ V$